INSTITUT UNIVERSITAIRE DES SCIENCES
Faculté *des* Sciences *et* Technologies

Licence III

Cours : SYSTEME I

PROFESSEUR: ISMAEL ST AMOUR

TD I

***Présenté par* :**

Raphaël LANGLOIS

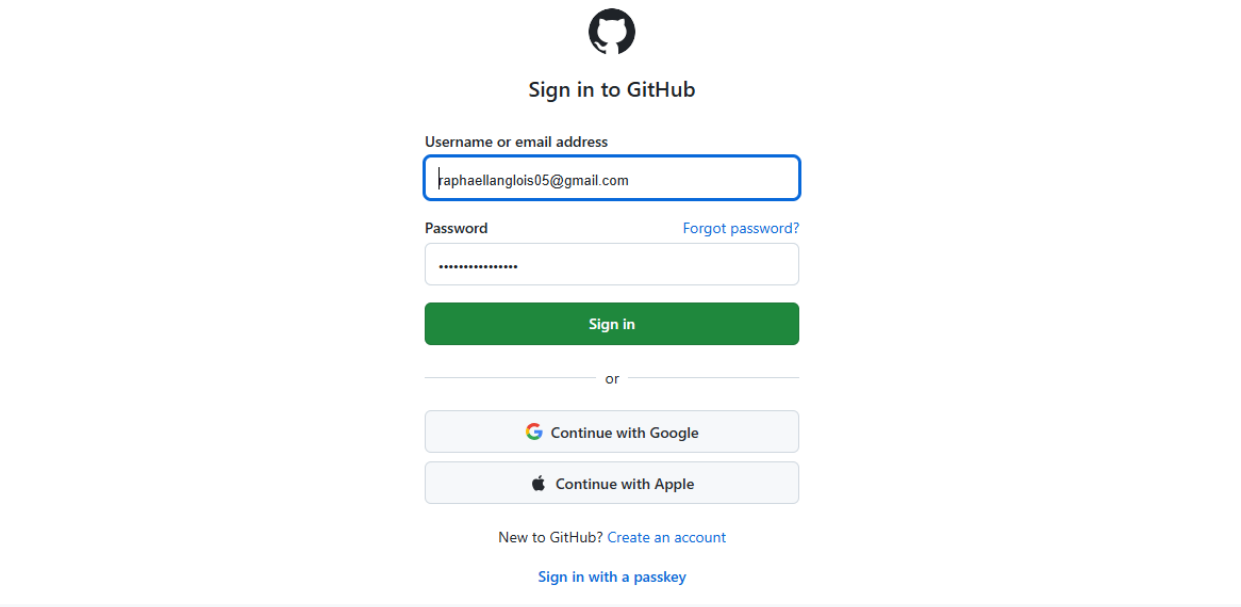
19 OCTOBRE 2026

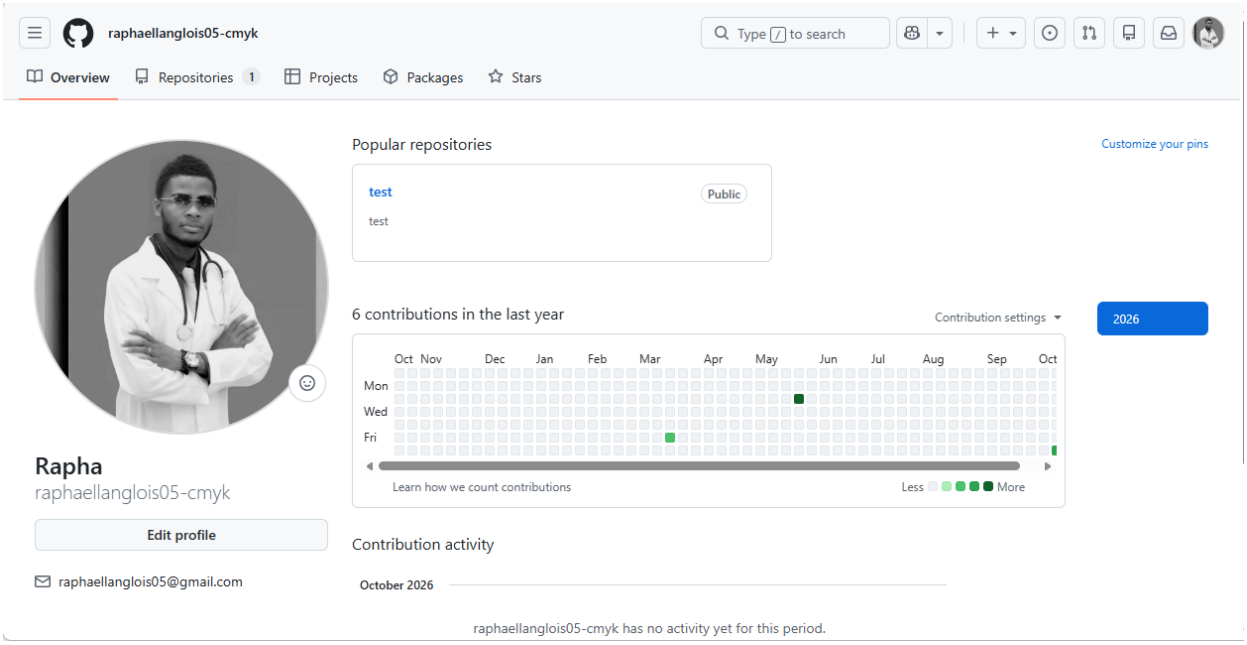
1) VERIFICATION DE LA VERSION DE GIT INSTALLEE

```
LENOVO YOGA@LENOVO UCRT64 ~
$ git version
git version 2.56.0.windows.2

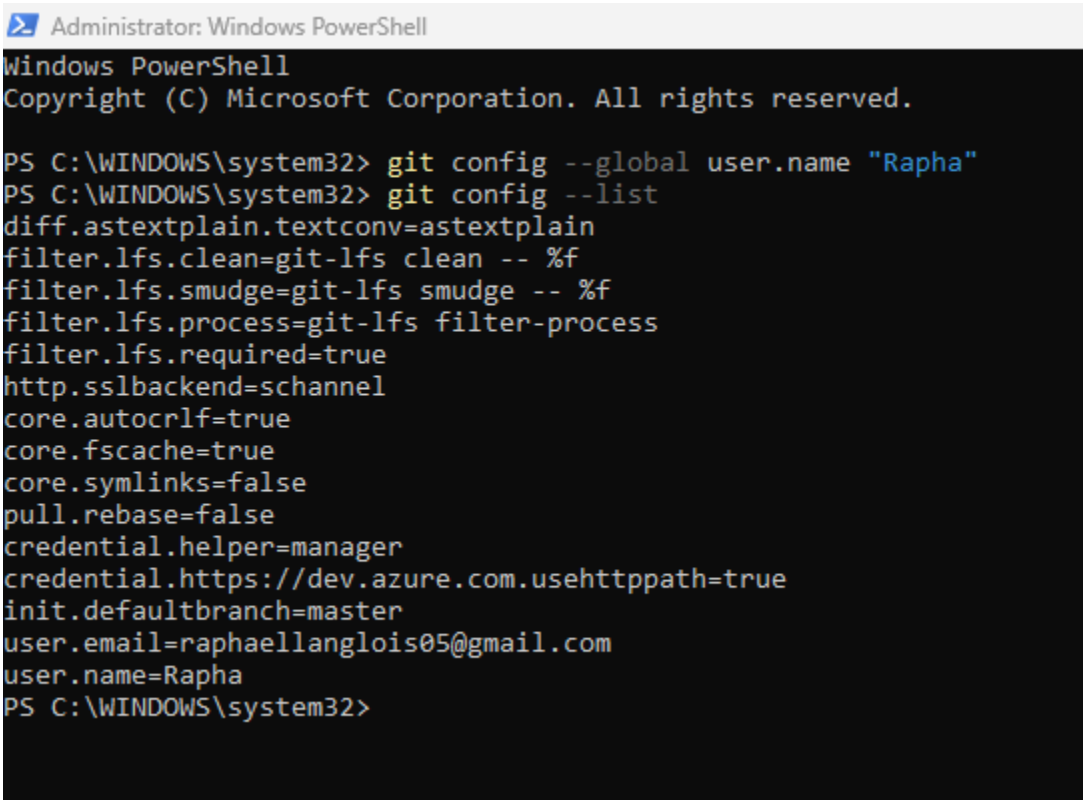
LENOVO YOGA@LENOVO UCRT64 ~
$
```

2) CONNEXION GIT-UB ONLINE

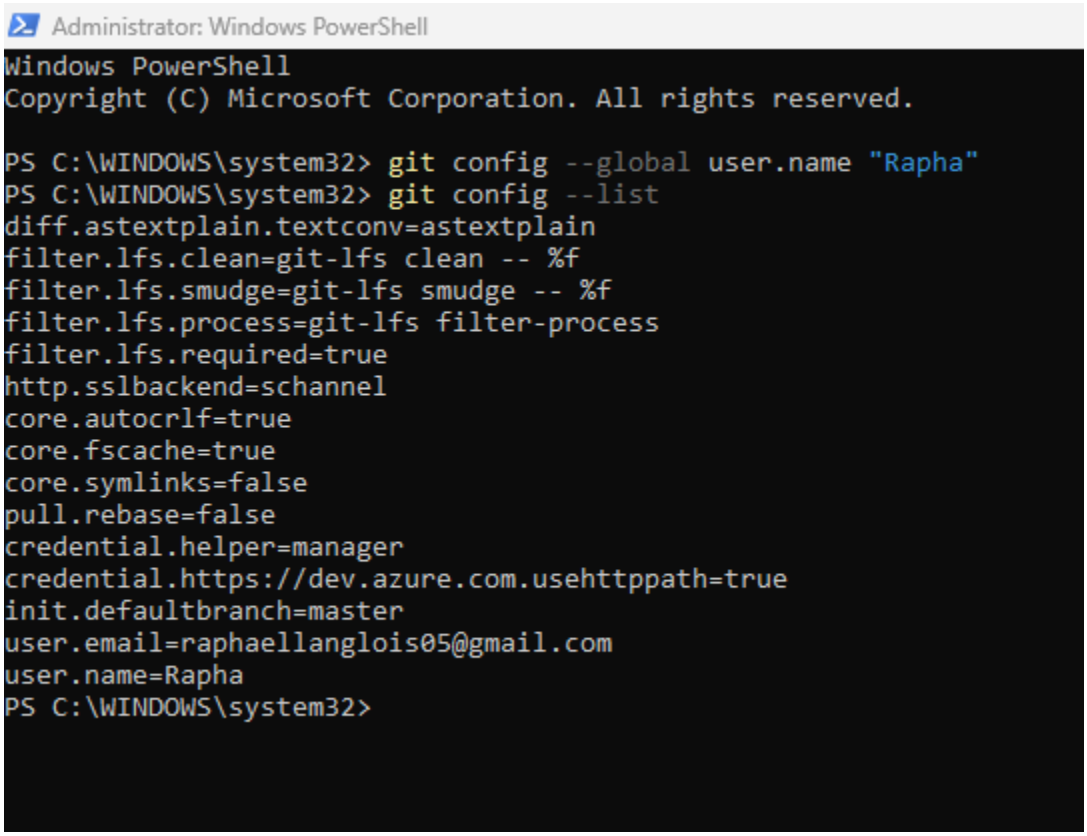




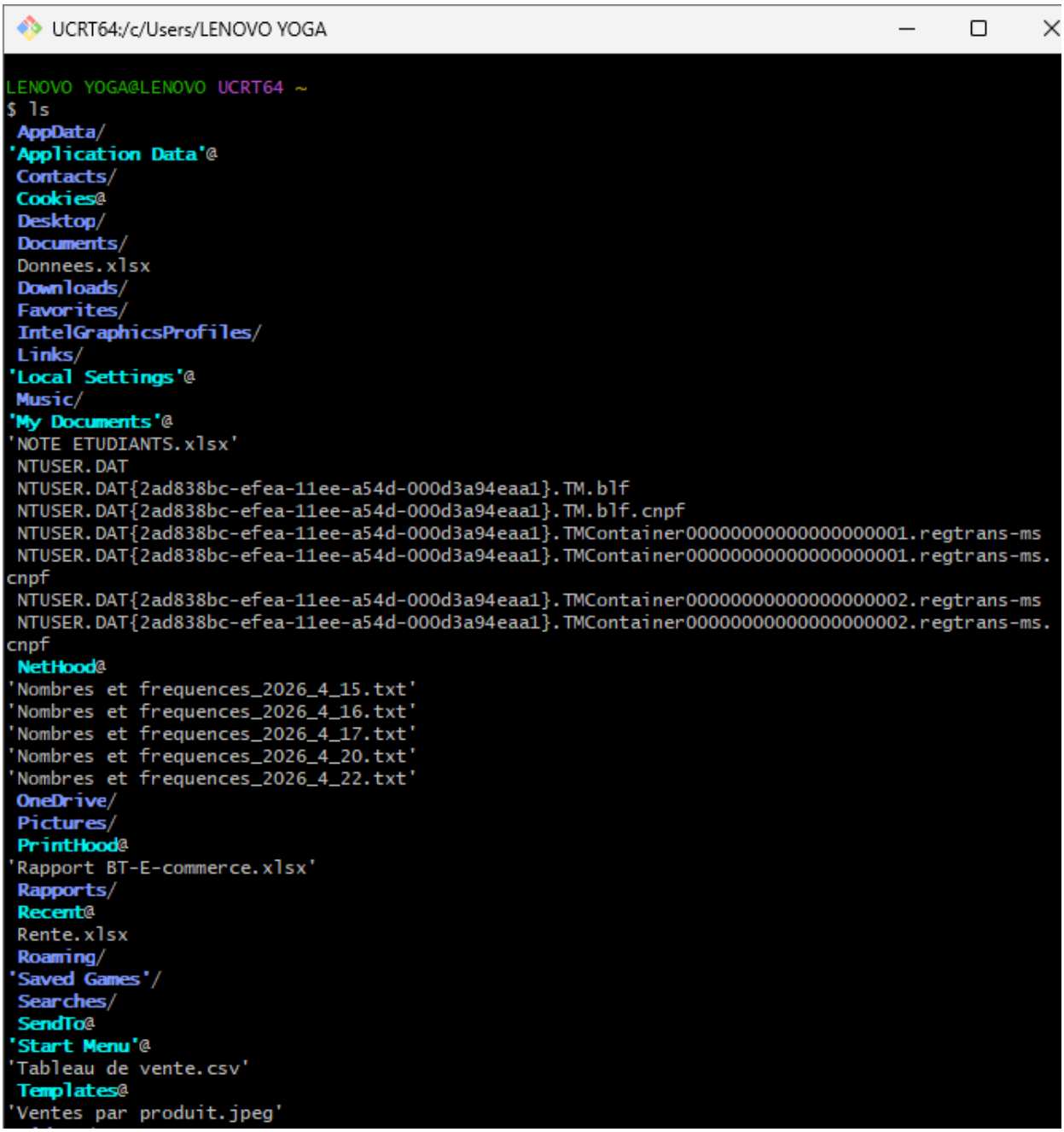
3) CONFIGURATION DE GIT AVEC W. POWERSHELL

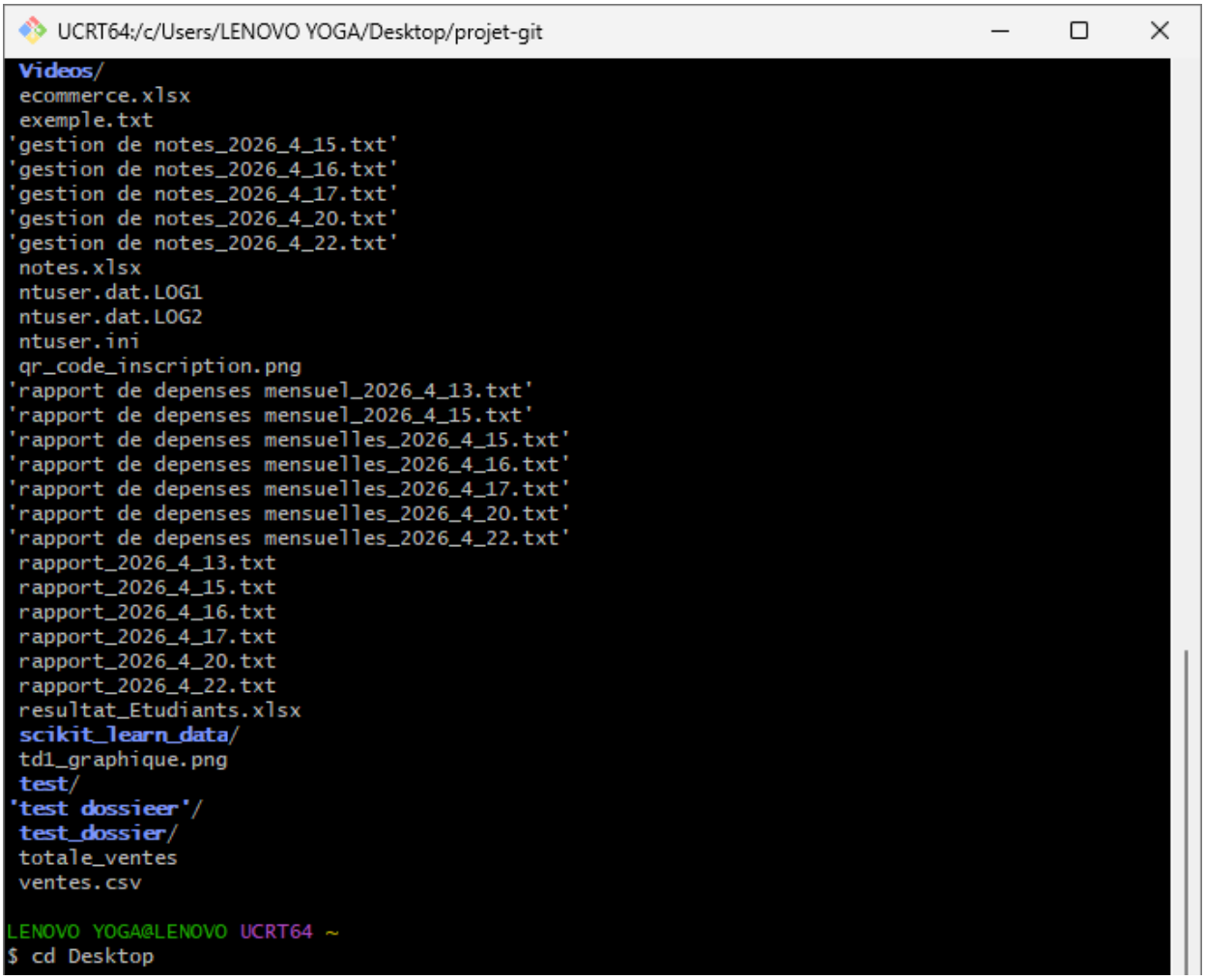


4) VERIFICATIONS DES DONNEES UTILISATEURS



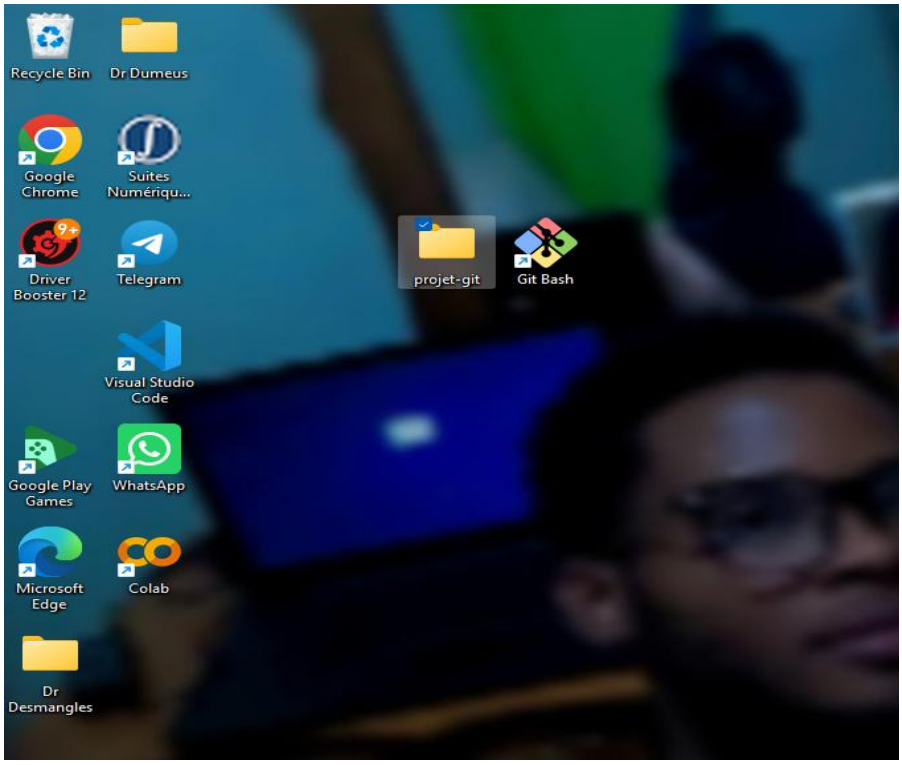
5) AFFICHAGE DE LA LISTE DES FICHIERS ET DOSSIERS AVEC LA COMMANDE LS





6) CREATION D'UN DOSSIER DANS DESKTOP





7) ENTREE DANS NOTRE DOSSIER « PROJET1-GIT » FRAICHEMENT CREE.

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop
$ cd projet-git

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git
$ git init
Initialized empty Git repository in C:/Users/LENOVO YOGA/Desktop/projet-git/.git/

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ echo "Hello  Git" > README.md

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git tou
ches it

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ git commit -m "Premier commit avec PowerShell"
[master (root-commit) de94b76] Premier commit avec PowerShell
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ git status
On branch master
nothing to commit, working tree clean

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$
```

8) INITIALISATION DU DEPOT GIT.

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop
$ cd projet-git

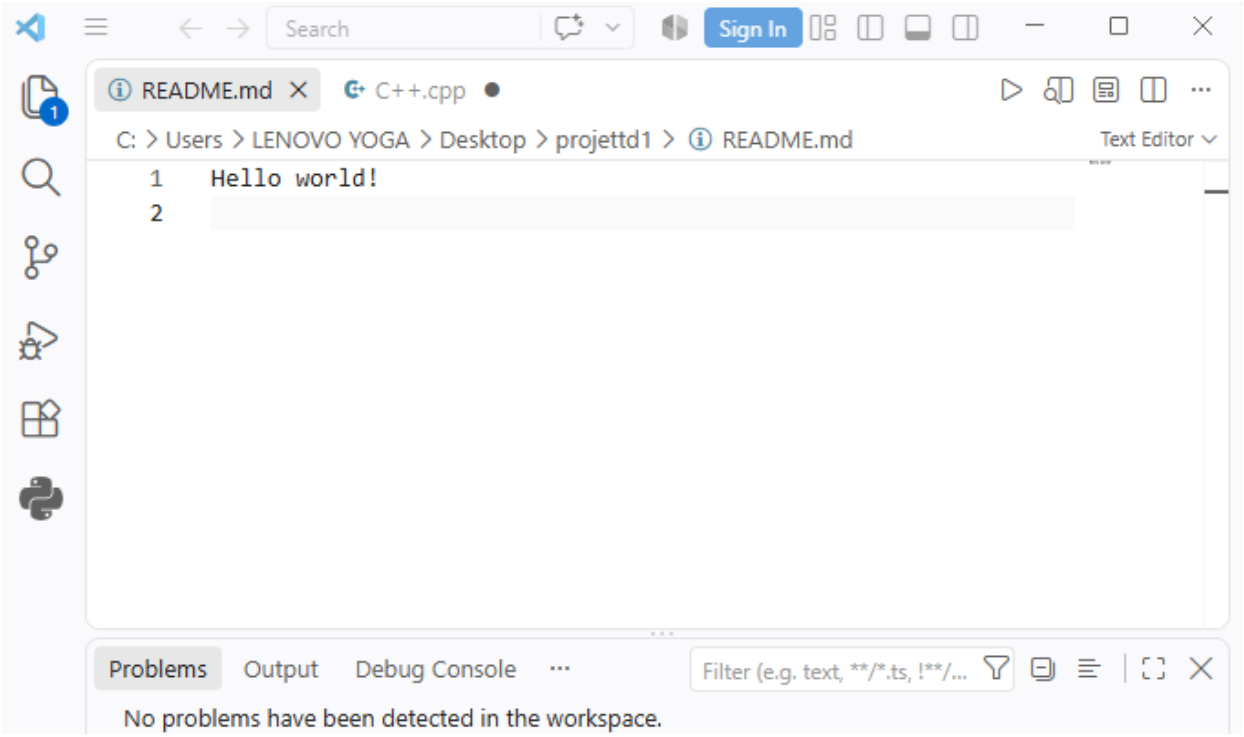
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git
$ git init
Initialized empty Git repository in C:/Users/LENOVO YOGA/Desktop/projet-git/.git/
```

9) OUVERTURE ET ECRITURE DE MON PROJET DANS LE FICHIER README.MD

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop
$ cd projet-git

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git
$ git init
Initialized empty Git repository in C:/Users/LENOVO YOGA/Desktop/projet-git/.git/

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ echo "Hello  Git" > README.md
```



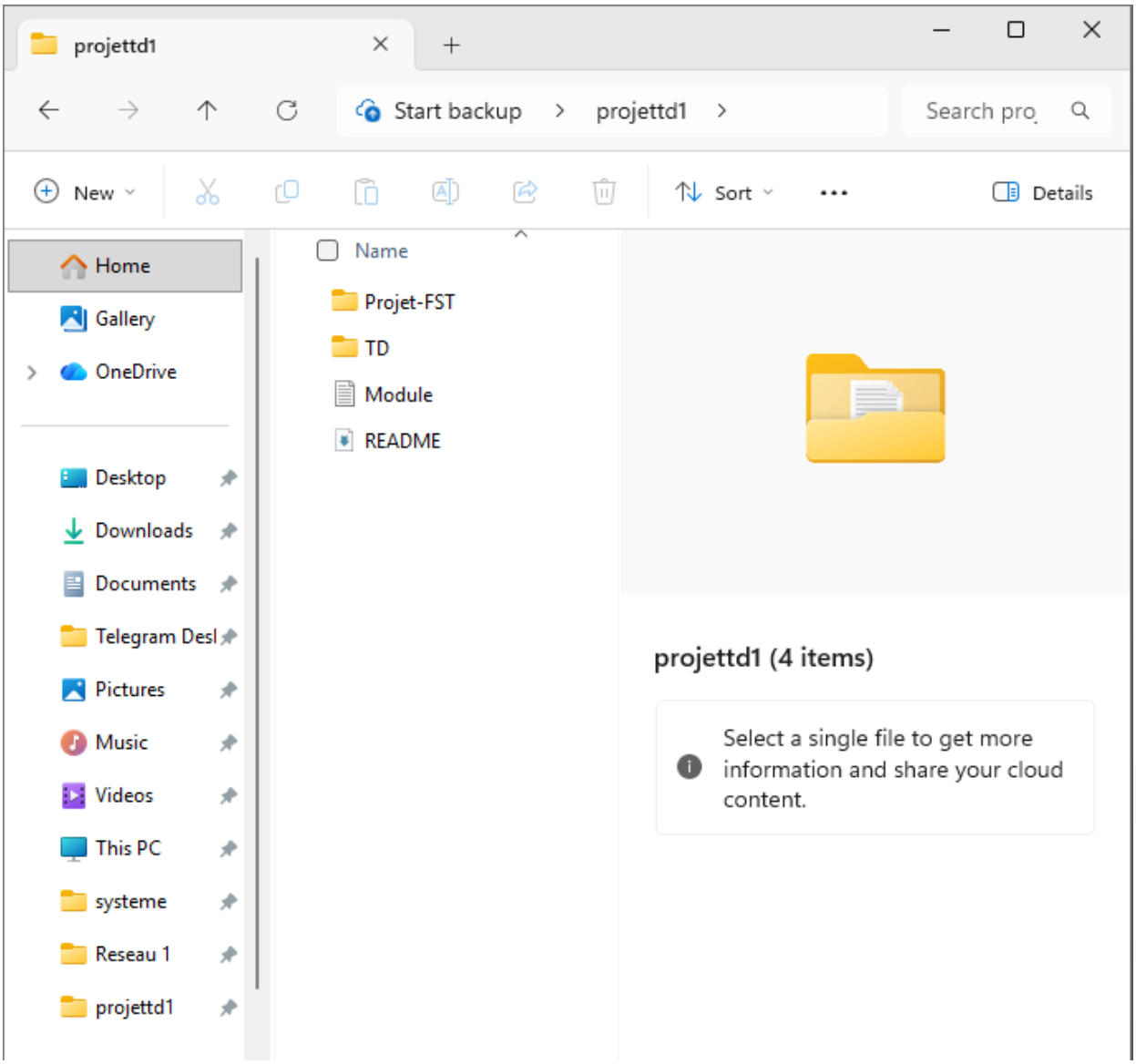
10) AJOUT DU DOSSIER README.MD DANS GIT ET PREMIER COMMIT EFFECTUE

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git tou
ches it

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ git commit -m "Premier commit avec PowerShell"
[master (root-commit) de94b76] Premier commit avec PowerShell
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$ git status
On branch master
nothing to commit, working tree clean

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projet-git (master)
$
```



11) AFFICHAGE DE L'HISTORIQUE DES COMMITS DANS UN DEPOT GIT.

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ git log
commit a974b18ede08dd07ab36db60767d823432a5f9be (HEAD -> main)
Author: Rapha <raphaellanglois05@gmail.com>
Date:   Sun Oct 18 00:52:27 2026 -0400

    Ajout module

commit 9b5efdf6312b6fb034e3a67edfe777608db35db8
Author: Rapha <raphaellanglois05@gmail.com>
Date:   Sun Oct 18 00:28:29 2026 -0400

    Premier commit

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$
```

12) GENERATION D'UNE CLE SSH

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ ssh-keygen -t rsa -b 4096 -C "raphaellanglois05@gmail.com"
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/LENOVO YOGA/.ssh/id_rsa):
/c/Users/LENOVO YOGA/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase for "/c/Users/LENOVO YOGA/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/LENOVO YOGA/.ssh/id_rsa
Your public key has been saved in /c/Users/LENOVO YOGA/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:fPl5TZmBJVsxEYpYtTHbsP6Ndx9+GFOyB3K73jWn/XO raphaellanglois05@gmail.com
The key's randomart image is:
+---[RSA 4096]----+
|           ..* B+|
|          o o ^ .|
|         . . X = |
|        .   .. + *|
|         S o  . Oo|
|          . . .+==|
|             o.oOO|
|              .=.E|
|                .B|
+----[SHA256]-----+
```

13) AFFICHAGE DE LA CLE PRIVEE AVEC LA COMMANDE CAT

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ cat ~/.ssh/id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAACFwAAAAdzc2gtcn
NhAAAAAwEAAQAAAgEApl2OyKfhoR6hja0PgHZXxSiBbONNI7wSNpx8neV58+ycKo1FT8ff
S4SOYX2OnaHkg5VJFXSq1G6lHJOAe1EaLVd7Iiwx/LLRg8lZ6gZJ1RS4Er4UUZCDpw/H78
w3W1oMNxpF62j9jJIhCnM7MB5T5Yu/ch/lrEe/eIhOwK2VeOJHG2iKhbJa8IoWxBRtjy17
sgUdjevos6pkOoFjHN6FKArN/dRujv+s5ZdSza2b1THxpYtDseafu4l+SmoiK4mYIo5vmH
X/bjdzA3w+BG+FIsENZiN9qZj5zirciADJhNPhwuYiDCa12Q7fpt3OYjZYq3oiODm3vuUG
uonlMCm2j9UnSvaqQEOWUBKe57aV7q1eVRHhwtG1IRkWIsSIP4Ql1ZbcirvJopbbo4i4wW
M9XbLiPbn5KGAd2HcICd8a9zKOAW7NhDMwvlVbR3LnnRbRgb8fdy+veBGZBPfMru9X4GS5
XtFxfM/U8oKNZUNP6t+KgtXP5BXT5fjb8O2LRI7YFkUorFMEzAMBQpaVxrN2MsTVE7qByN
6SAtduINAqVmnLoTQGZosU3UIyzPJMrHG7qs+9sKybVOc64ymqVzl5hRQutI8KWU9WwTiS
i7yxB7IgVcnz1a9lvmeKCVRon9Q41MmQ+wIu1GhwXlh4fuaEbcX4/OmlbkCM1U7zocPxz+
EAAAdQaVXKt2lVyrcAAAAHc3NoLXJzYQAAAgEApl2OyKfhoR6hjaOPgHZXxSiBbONNI7wS
Npx8neV58+ycKo1FT8ffS4SOYX2OnaHkg5VJFXSq1G6lHJOAe1EaLVd7Iiwx/LLRg8lZ6g
ZJ1RS4Er4UUZCDpw/H78w3W1oMNxpF62j9jJIhCnM7MB5T5Yu/ch/lrEe/eIhOwK2VeOJH
G2iKhbJa8IoWxBRtjy17sgUdjevos6pkOoFjHN6FKArN/dRujv+s5ZdSza2b1THxpYtDse
afu4l+SmoiK4mYIo5vmHX/bjdzA3w+BG+FIsENZiN9qZj5zirciADJhNPhwuYiDCa12Q7f
pt3OYjZYq3oiODm3vuUGuonlMCm2j9UnSvaqQEOWUBKe57aV7q1eVRHhwtG1IRkWIsSIP4
Ql1ZbcirvJopbbo4i4wWM9XbLiPbn5KGAd2HcICd8a9zKOAW7NhDMwvlVbR3LnnRbRgb8f
dy+veBGZBPfMru9X4GS5XtFxfM/U8oKNZUNP6t+KgtXP5BXT5fjb8O2LRI7YFkUorFMEzA
MBQpaVxrN2MsTVE7qByN6SAtduINAqVmnLoTQGZosU3UIyzPJMrHG7qs+9sKybVOc64ymq
Vzl5hRQutI8KWU9WwTiSi7yxB7IgVcnz1a9lvmeKCVRon9Q41MmQ+wIu1GhwXlh4fuaEbc
X4/OmlbkCM1U7zocPxz+EAAAADAQABAAACAB1ehoczx+urHicGKanqemUmWO7Rk7EexvJY
HtG7DJPorcr/RICojr1ksHlTeTqotVAbXxsIDQ5SeBh626ApeZ8OhgDK34o2QTFsLtDXCl
qw/Svl3W1mvy8TZJ7rJMyeMXDa1QQRdU1HEMko/YZpnDL2QDZT9jY7Nkx/d7OhETYAxc/L
XUTr7wzuu8UivBKeG2uaZpTC1EI/buwQhmA2Ojf27FWGlnvtG+cuFyKTJuLOna21T56M0Z
8ITOCwytZ3xGE5Vu8Jya154XZ4jPQ4gb7YaNaiplD13ges3mCuDuVBiGXhPZp6jPjEHOx2
22kpIe8nsRSkU9FB2jW9GTYu7vjdAi+4UYAdJgbJwPhg9cnZ2wf6vBOOOoAYlKzeEcgOxx
46QFTiDGMq5Pb6ZbLR9FimHymN/1Uz8EoeIGOYdqPuEo/9c1NYpGZKL3xVuhS+9YEAvsJG
YyJ+KNH/KtNUFPHTODcf8fNIRTK/er+5/XRfL9dtgs4t9jR4dO7Np1iOYi/ClQuNinwyHm
LNwDxe3gb5LU4NCkwCyp6GIH2HBkU8rUn3VbnJFnNTbPKgEwMjuzzwJYO7xB6uymUpQ+UK
x2cN+4ZrAenO5+NwPzXi2eOPv29UtsVsPo37ObtWZiuzlNJKwDb/WDvYUoY7Ba6jOGlqul
VVBXggyA8R5z/UEdTnAAABAD/zVXl3GGpS7esW6ilMvrL/OYIIOOr3K+XIJpyl2PBd9T1X
eO4eaOhut1LSXlGGwOm4GaODIrxRGBCtN528tk5Kdb7vlVryavCWxto+41pAx/vMW7qUM1
QUt8tQQA4OUVKO7Cv2//lFQ6XHRJ3NbQsR2IyQw8zerTKtHs/hHeui1ueh34UH6KHEjanw
s1kh/KoBPnrjzWkP2JwzeVSd5M+Vfhuc/UTspxGd6NS/GNvLyj7C/Zhm4vPNjKZ1/GXqRh
SIV7JZgSr8VKzUyODETmf7beYOZfDARnSKfQZDy6EsydSGHeYFHZKFI5gDsMHWCRvnk/P5
Ev3tNUIENKPdnNAAAAEBAOYsOTEnxRIhSOZYxwbeOco6iaKKe5OzbaN/IbCnvqYfvlgDUK
uU1njfxGeB2HNk/K9EDhGcJibSrUQKhEmksTdD/4sVPwXoxUAoR2LsihIPj/CyKkwPBExF
ACIsGX+NsPOjLypccVSumS+YDW7IhmkWxVFE5kGjrH1KIcqiqMH34flZM96CJGOiqYiG9r
tKdKqRCZsOlHxSFOc7UMAFiQTEin8+IOTknXMsWMfKlzPORML4bU+rYtdOlUGZfaEfqSW6
21QdDp3sBWu+lzLHATy99nFZFOy7NRGQXrC2p19oCJt1tpxrdTZybGOHY+YvhVSIi9hW/E
VYN4bURYMRKq8AAAEBALkIngNk+eanOY+yUL+CpO++EZrzyO5/zi/nzXjDKvI9puXXPlNP
184aGaA7QMoSDRNVFdIY/EP+BW4Uc6FOTZPbHaXSvSD77GnDWKH3abCRwKJjXWSNdW9Vzs
cOTf18aRDxeoQsyxYP94d6KvN4PQdsqDZq+vrH2jBF5W1zoCHTBj5chzCcxT6b/F6cYRDd
9Uh+ieZEChugxlcIXpVrhqTuKMgEgFO8cuGBomwfhiDthRuecZdsMFY3TwgMo5cJIcRrlh
```

```
LNwDxe3gb5LU4NCkwCyp6GIH2HBkU8rUn3VbnJFnNTbPKgEwMjuzzwJYO7xB6uymUpQ+UK
x2cN+4ZrAenO5+NwPzXi2eOPv29UtsVsPo37ObtWZiuzlNJKwDb/WDvYUoY7Ba6jOGlqul
VVBXggyA8R5z/UEdTnAAABAD/zVXl3GGpS7esW6ilMvrL/OYIIOOr3K+XIJpyl2PBd9T1X
eO4eaOhut1LSXlGGwOm4GaODIrxRGBCtN528tk5Kdb7vlVryavCWxto+41pAx/vMW7qUM1
QUt8tQQA4OUVKO7Cv2//lFQ6XHRJ3NbQsR2IyQw8zerTKtHs/hHeui1ueh34UH6KHEjanw
s1kh/KoBPnrjzWkP2JwzeVSd5M+Vfhuc/UTspxGd6NS/GNvLyj7C/Zhm4vPNjKZ1/GXqRh
SIV7JZgSr8VKzUyODETmf7beYOZfDARnSKfQZDy6EsydSGHeYFHZKFI5gDsMHWCRvnk/P5
Ev3tNUIENKPdnNAAAAEBAOYsOTEnxRIhSOZYxwbeOco6iaKKe5OzbaN/IbCnvqYfvlgDUK
uU1njfxGeB2HNk/K9EDhGcJibSrUQKhEmksTdD/4sVPwXoxUAoR2LsihIPj/CyKkwPBExF
ACIsGX+NsPOjLypccVSumS+YDW7IhmkWxVFE5kGjrH1KIcqiqMH34flZM96CJGOiqYiG9r
tKdKqRCZsOlHxSFOc7UMAFiQTEin8+IOTknXMsWMfKlzPORML4bU+rYtdOlUGZfaEfqSW6
21QdDp3sBWu+lzLHATy99nFZFOy7NRGQXrC2p19oCJt1tpxrdTZybGOHY+YvhVSIi9hW/E
VYN4bURYMRKq8AAAEBALkIngNk+eanOY+yUL+CpO++EZrzyO5/zi/nzXjDKvI9puXXPlNP
184aGaA7QMoSDRNVFdIY/EP+BW4Uc6FOTZPbHaXSvSD77GnDWKH3abCRwKJjXWSNdW9Vzs
cOTf18aRDxeoQsyxYP94d6KvN4PQdsqDZq+vrH2jBF5W1zoCHTBj5chzCcxT6b/F6cYRDd
9Uh+ieZEChugxlcIXpVrhqTuKMgEgFO8cuGBomwfhiDthRuecZdsMFY3TwgMo5cJIcRrlh
WrJLoScyGuqDAJsOwV3bjVTGHh7T4g5P8zqqQAUYRE+3+RwjOy2wKKdxKKigBaWwz21y9u
ubqpgY2AEm8AAAAbcmFwaGFlbGxhbmdsb2lzMDVAZ21haWwuY29t
-----END OPENSSH PRIVATE KEY-----

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$
```

14) AFFICHE DE LA CLE PUBLIQUE

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQCmXbTIp+GhHqGNrQ+AdlfFKIFs4OOjvBI2nHyd5Xnz7JwqjUVPx99LhL
RhfbSdoeSDlUkVdKrUbqUck4B7URotV3siLDH8stGDyVnqBknVFLgSvhRRkIOnD8fvzDdbWgw3GkXraP2MkiEKczswHlPl
i79yH+WsR794iE7ArZV7QkcbaIqFslrwihbEFG2PLXuyBR2N6+izqmTSgWMc3oUoCs391G6O/6zll1LNrZvVMfGliOOx5p
+7iX5KaiIriZgijm+Ydf9uN3MDfD4Eb4UiwQ1mI32pmPnOKtyIAMmEO+HC5iIMJrXZDt+m3fRiNlireiLQObe+5Qa6ieUw
KbaP1SdK9qpAQ5ZQEp7ntpXurV5VEeHCObUhGRYixIg/hCXVltyKu8miltujiLjBYz1dsuI9ufkoYB3YdwgJ3xr3Mo4Bbs
2EMzC+VVtHcuedFtGBvx93L694EZkE98yu71fgZLleOXF8z9Tygo1lQO/q34qC1c/kFdPl+Nvw7YtEjtgWRSisUwTMAwFC
lpXGs3YyxNUTuoHI3pIC124gOCpWacuhNAZmixTdQjLM8kyscbuqz72wrJtXRzrjKapXOXmFFC6OjwpZT1bBOJKLvLEHsi
BVyfPVr2W+Z4oJVGif1DjUyZD7Ai7UaHBeWHh+5oRtxfj/SaVuQIzVTvOhw/HP4Q== raphaellangloisO5@gmail.com

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$
```

15) AJOUT REUSSIE DE LA CLE SUR MON COMPTE GITHUB



Rapha (raphaellanglois05-cmyk)
Your personal account

Menu

SSH keys

New SSH key

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

key Win
SHA256:jy5N3HCEpb4bX2A0w3cTzyLOGdOVyxU0inXXRcmBKo4
SSH
Added on Oct 18, 2026
Last used within the last week — Read/write
Delete

Check out our guide to connecting to GitHub using SSH keys or troubleshoot common SSH problems.

GPG keys

New GPG key

[GitHub] A new SSH authentication public key was added to your account

Inbox ×

GitHub <noreply@github.com> 12:36 AM (22 hours ago)
to me

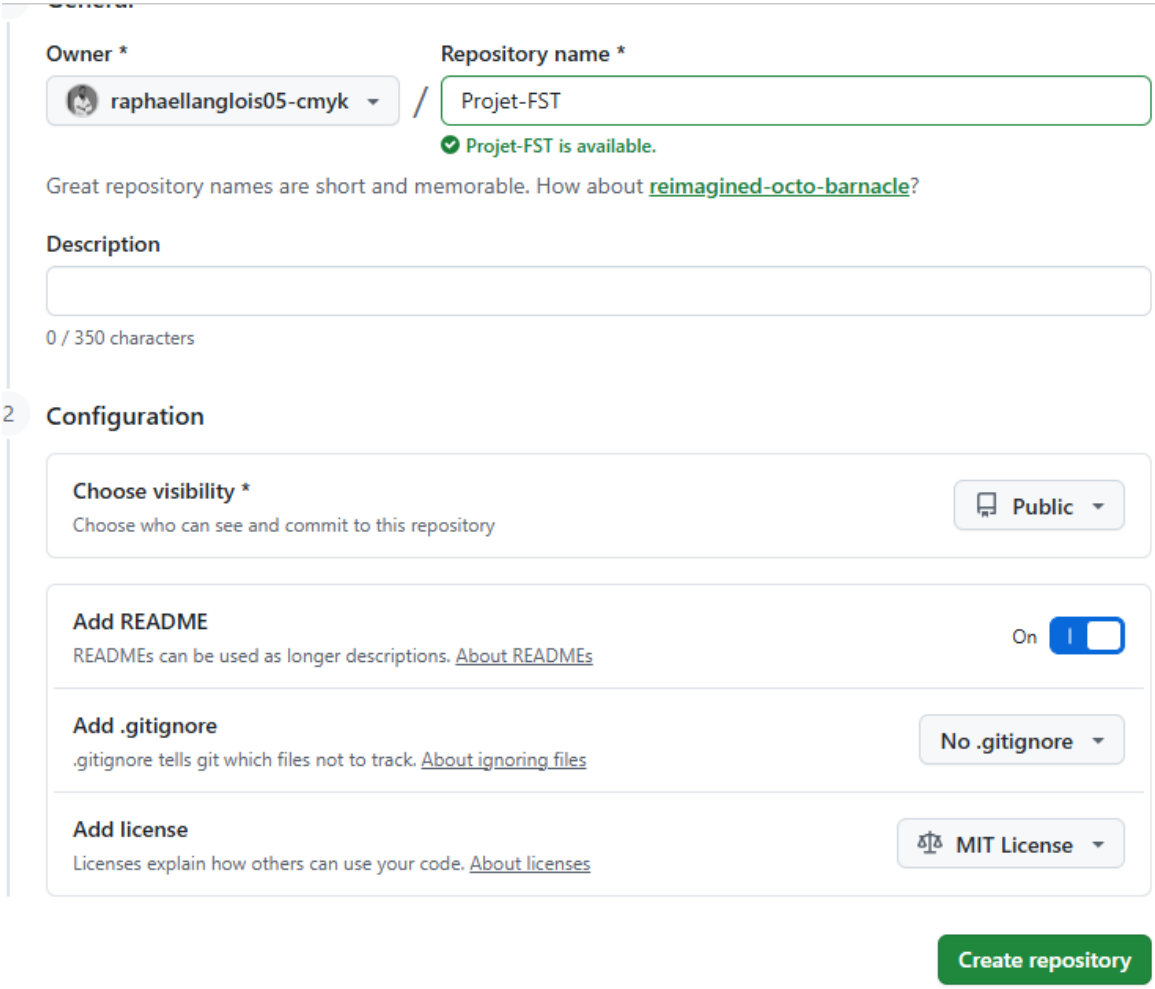
The following SSH key was added to your account:

key Win
SHA256:jy5N3HCEpb4bX2A0w3cTzyLOGdOVyxU0inXXRcmBKo4

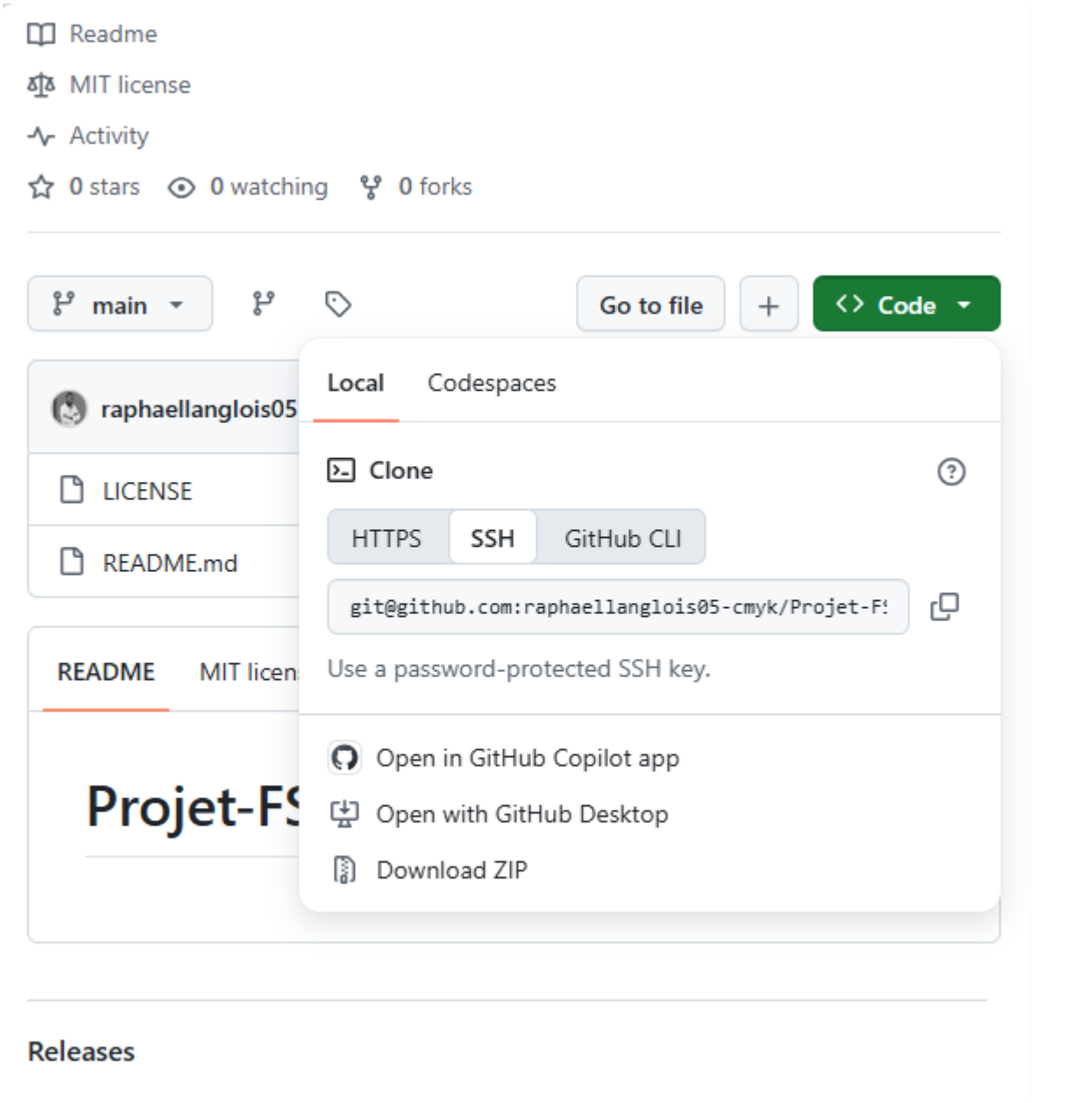
If you believe this key was added in error, you can remove the key and disable access at the following location:

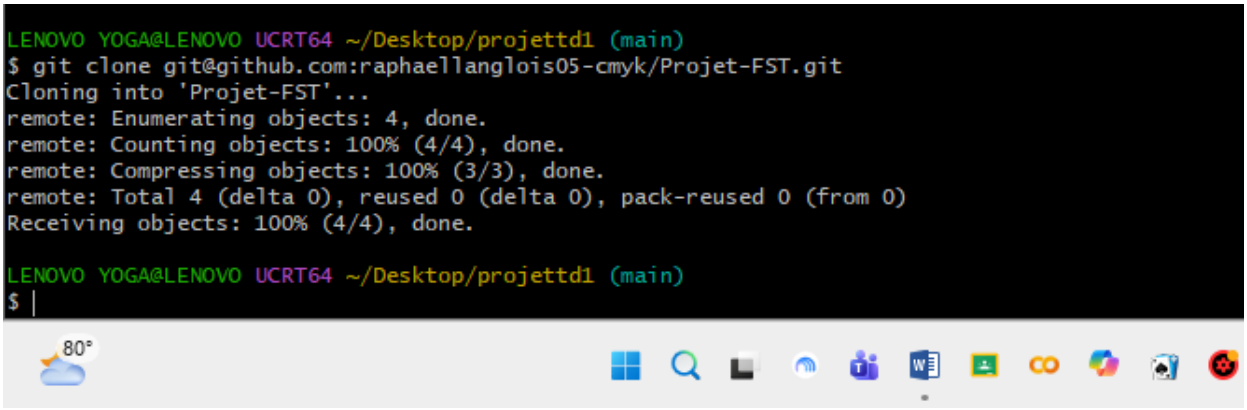
https://github.com/settings/keys

16) CREATION DU DEPOT GITHUB



17) CLONAGE DU DEPOT GITHUB VIA LA CLE SSH


Readme
MIT license
Activity
0 stars
0 watching
0 forks
main
Go to file
Code
raphaellanglois05
LICENSE
README.md
README
Local
Codespaces
Clone
HTTPS
SSH
GitHub CLI
git@github.com:raphaellanglois05-cmyk/Projet-F
Use a password-protected SSH key.
Open in GitHub Copilot app
Open with GitHub Desktop
Download ZIP
Releases


LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ git clone git@github.com:raphaellanglois05-cmyk/Projet-FST.git
Cloning into 'Projet-FST'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$
80°

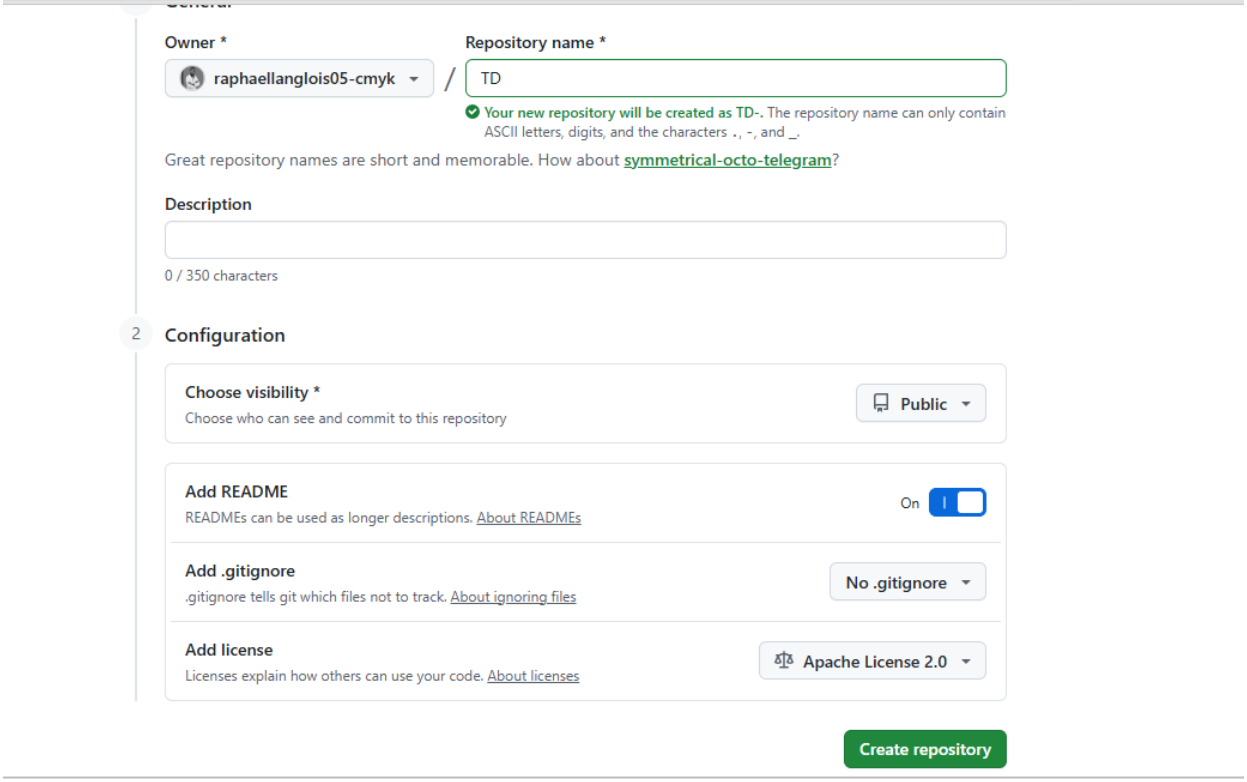
```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ echo "Systeme d'exploitation" > Module.txt

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ git add .
warning: in the working copy of 'Module.txt', LF will be replaced by CRLF the next time Git to
uches it

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ git commit -m "Ajout Module"
[main 711327f] Ajout Module
 1 file changed, 1 insertion(+)
 create mode 160000 Projet-FST

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ git branch -M main
```

SECOND EXERCICE



```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ git clone git@github.com:raphaellanglois05-cmyk/TD.git
Cloning into 'TD'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$
```

```
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1 (main)
$ cd TD

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD (main)
$ mkdir systeme

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD (main)
$ cd systeme

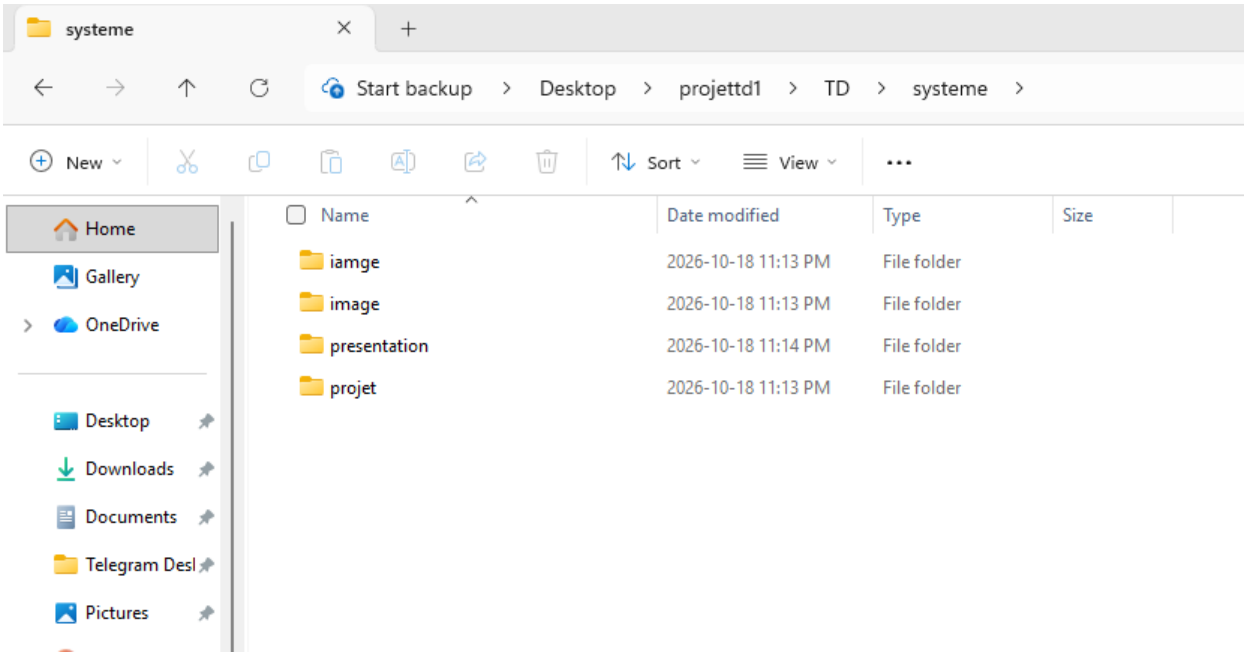
LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD/systeme (main)
$ mkdir iamge

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD/systeme (main)
$ mkdir image

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD/systeme (main)
$ mkdir projet

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD/systeme (main)
$ mkdir presentation

LENOVO YOGA@LENOVO UCRT64 ~/Desktop/projettd1/TD/systeme (main)
$
```


systeme
Start backup
Desktop
projettd1
TD
systeme
New
Sort
View
Home
Gallery
OneDrive
Desktop
Downloads
Documents
Telegram Desl
Pictures
Name
Date modified
Type
Size
iamge
2026-10-18 11:13 PM
File folder
image
2026-10-18 11:13 PM
File folder
presentation
2026-10-18 11:14 PM
File folder
projet
2026-10-18 11:13 PM
File folder